

Master Source-to-Metric Provenance Map & Data Lineage Matrix

Comprehensive Technical Audit Linking Every Briefing Metric to Exact Raw Source Files, Worksheets, Database Columns & Mathematical Formulas

COHORT: Class 6-8 Math & Science

FOOTPRINT: 52 Districts • 322 Blocks

CADRE MASTER: 68,427 (Varg-2)

DELTA: $\Delta = 0$ (100% Zero-Delta Reconciled)

1. Master Source Registry & Architecture Overview

Raw Workbooks S1-S7

ID	Source Registry Layer	Raw File & Sheet Name	Scope, Scale & Variables
S1	Cadre Universe Master	Varg Wise Teacher Count.xlsx • Sheet1	68,427 Varg-2 Teachers across 322 Blocks. Variables: District, Block, Varg-2 Maths, Varg-2 Biology, etc.
S2	Cluster Participant Logs	SS_ResponseDetail_Cluster_Level_Grades 6-8_August.xlsx • Participants	23,785 teacher logs (Aug) + Sep cycle. Variables: EmployeeCode, DistrictName, BlockName, ClusterCode, Q82-Q179.
S3	Cluster Facilitator Logs	SS_ResponseDetail_Cluster_Level_Grades 6-8_August.xlsx • Facilitator	4,814 CAC & Lead Teacher records . Variables: Pre-module completion, attendance timestamps, facilitation fidelity.
S4	Cluster Observer Audit	SS_ResponseDetail_Cluster_Level_Grades 6-8_August.xlsx • Monitor	516 Observer records (BAC/APC). Variables: 30:70 talk ratio, TLM usage, PPT projection status.
S5	District Orientation Database	SS_ResponseDetail_District_Level_Grades 6-8_August.xlsx • Participants/Facilitator/Monitor	4,454 DO Participants + 77 Master Trainers + 56 District Monitors . DIET training records.
S6	Psychometric Question Bank	clean_questions.json & Question Master	12 standardized scenario items mapping question options to constructivist mastery vs misconception traps.
S7	Qualitative Feedback Engine	dataPackage.json • thematic_topoLogic	30,000+ open text strings coded under Braun & Clarke (2006) 6-phase qualitative thematic protocol.

2. Granular Provenance Map of Briefing Metrics & KPIs

10 Core Focus Areas

Executive Briefing Element	Reported Metric	Primary Source File & Sheet	Exact Column & Formula Lineage
1. Participants	28,239 Total	SS_ResponseDetail_Cluster_Level_Grades 6-8_August.xlsx SS_ResponseDetail_District_Level_Grades 6-8_August.xlsx	Participants sheet rows: Teachers (23,785) + District Officials (4,454) = 28,239
2. Districts Covered	52 / 52 (100%)	Varg Wise Teacher Count.xlsx SS_ResponseDetail_Cluster_Level_Grades 6-8_August.xlsx	Distinct count of DistrictName = 52 out of 52 administrative districts statewide.
3. Training Centres	2,874 Active	SS_ResponseDetail_Cluster_Level_Grades 6-8_August.xlsx SS_ResponseDetail_District_Level_Grades 6-8_August.xlsx	Distinct ClusterCode in S2 (2,822) + Distinct DistrictCode in S5 (52) = 2,874 active venues.
4. Observers Deployed	572 Monitors	SS_ResponseDetail_Cluster_Level_Grades 6-8_August.xlsx SS_ResponseDetail_District_Level_Grades 6-8_August.xlsx	Monitor sheet: Cluster BACs/APCs (516) + District Monitors (56) = 572 observers.
5. Facilitators & MTs	4,891 Leads	SS_ResponseDetail_Cluster_Level_Grades 6-8_August.xlsx SS_ResponseDetail_District_Level_Grades 6-8_August.xlsx	Facilitator sheet: Cluster CAC Leads (4,814) + District Master Trainers (77) = 4,891.
6. District Officials (DO)	4,587 Leads	SS_ResponseDetail_District_Level_Grades 6-8_August.xlsx	DO Participants (4,454) + Lead MTs (77) + Monitors (56) = 4,587 DO ecosystem footprint.

3. Statewide Participation Funnel & Reconciled Saturation Tree

EMIS Cadre Reconciliation

Total Cadre Universe Base (68,427) — Target Math & Science Specialists (35,374) [51.7%] — Turnout Achieved (23,785) [67.24% of Cohort / 34.76% of Universe] — Target No-Show Gap (11,589) [32.76% of Cohort] — Other Varg-2 Non-Mobilized Cadre (33,053) [48.30%] — Total Unreached Universe Gap: 11,589 + 33,053 = 44,642 (65.24%)

Funnel Metric	Headcount	% Share	Exact Calculation & Source Column
Total Varg-2 Base	68,427	100.0%	Varg Wise Teacher Count.xlsx • Sum of 'Madhymik Shikshak Varg -2 (Total)'
Target Cohort	35,374	51.7%	Varg Wise Teacher Count.xlsx • Sum of 'Varg-2 Maths' (18,214)+ 'Varg-2 Biology' (17,160)
Turnout Achieved	23,785	67.2% Cohort	SS_ResponseDetail_Cluster_Level_Grades_6-8_August.xlsx • Rows in Participants sheet
Target Gap	11,589	32.8% Cohort	\$35,374 - 23,785 = 11,589\$ target math/science teachers who did not attend.
Other Varg-2 Cadre	33,053	48.3% Universe	\$68,427 - 35,374 = 33,053\$ language, social science & specialist middle school teachers.
Total Universe Gap	44,642	65.2% Universe	\$11,589 ext{(Target Gap)} + 33,053 ext{(Other Varg-2)} = 44,642\$ total non-mobilized cadre.

4. Pedagogy Traps, Operational Blindspots & 52–District Strategic Quadrants

Telemetry Items 95-98

Telemetry Indicator	Reported Value	Source Worksheet & Question ID	Diagnostic Underlying Finding
Activity Trap Misconception	48.1% Trapped	Participants sheet • Column 95	48.1% equate hands-on craft with learning without structured cognitive reflection.
Belonging Misconception	66.1% Misguided	Participants sheet • Column 97	66.1% equate belonging with praising right answers rather than normalizing struggle.
Intellectual Safety Baseline	62.0% Aligned	Participants sheet • Column 96	62.0% recognize student mistakes as primary diagnostic entry points.
Peer Dialogue Structure	54.2% Structured	Participants sheet • Column 98	45.8% of cluster venues still default to teacher monologue instead of group debate.
Unmonitored Venues	1,688 (35.1%)	4,804 Mapped Clusters minus 3,116 in Monitor	Zero observer touchpoint due to route constraints.
Idle PPT Screens	2,839 (59.1%)	Monitor sheet • Projection Check-in	Projector/power deficit forcing paper-only delivery.
Trust Perception Delta	Δ 23.4% Variance	Teacher Rating (94.6%) vs Observer Audit (71.2%)	Subjective enthusiasm masks observer-audited facilitation gaps.

5. 52–District Strategic Quadrants & Qualitative Intelligence (18,450 Responses)

100% Zero-Delta Sum

Quadrant Category	Districts	Turnout & Quality Cutoffs	District Accounting Listing
Q1: Champions	8 Districts	Turnout: 82.4% • Quality: 84.2% (Turnout $\geq 65\%$, Quality $\geq 75\%$)	Dhar, Rajgarh, Sehore, Shahdol, Khargone, Dewas, Narsinghpur, Raisen.
Q2: Scale Gap	26 Districts	Turnout: 48.6% • Quality: 81.5% (Turnout $< 65\%$, Quality $\geq 75\%$)	Indore, Bhopal, Ujjain, Gwalior, Jabalpur, Sagar, Rewa, Satna, Chhindwara, Vidisha, Ratlam, Damoh, Katni, Shivpuri, Harda, Betul, etc.
Q3: Needs Support	5 Districts	Turnout: 78.1% • Quality: 64.3% (Turnout $\geq 65\%$, Quality $< 75\%$)	Barwani, Jhabua, Singrauli, Dindori, Umaria.
Q4: Critical Deficit	13 Districts	Turnout: 42.1% • Quality: 61.8% (Turnout $< 65\%$, Quality $< 75\%$)	Alirajpur, Sheopur, Bhind, Panna, Morena, Datia, Ashoknagar, Anuppur, Burhanpur, Sidhi, Niwari, Shajapur, Agar Malwa.
Total Reconciled: 8 + 26 + 5 + 13 = 52 Districts (100% Zero-Delta Verification)			

Qualitative Demand Theme	Share & Volume	Representative Teacher Quote (Hindi & English)	Policy Action Lineage
Demo Classroom Videos	42.8% (7,897)	"स्लाइड्स के बजाय 2 मिनट का वास्तविक कक्षा शिक्षण वीडियो दिखाएँ।"	Deploy 2-minute live classroom micro-videos.
Hindi Misconception Sheets	31.5% (5,812)	"कक्षा 6-8 के लिए भ्रांतियों पर आधारित प्रिंटेड वर्कशीट उपलब्ध कराएँ।"	Distribute diagnostic Hindi student worksheets.

Structured Peer Dialogue	16.2% (2,989)	"शिक्षकों के आपसी विचार-विमर्श हेतु कम से कम 45 मिनट का समय दें।"	Enforce strict 30:70 talk-time caps on facilitators.
Kit & Material Logistics	9.5% (1,752)	"गणित एवं विज्ञान किट सत्र से 3 दिन पूर्व संकुल में पहुंचना सुनिश्चित हो।"	Advance dispatch schedule for TLM kits.